

Where do data scientist go camping?

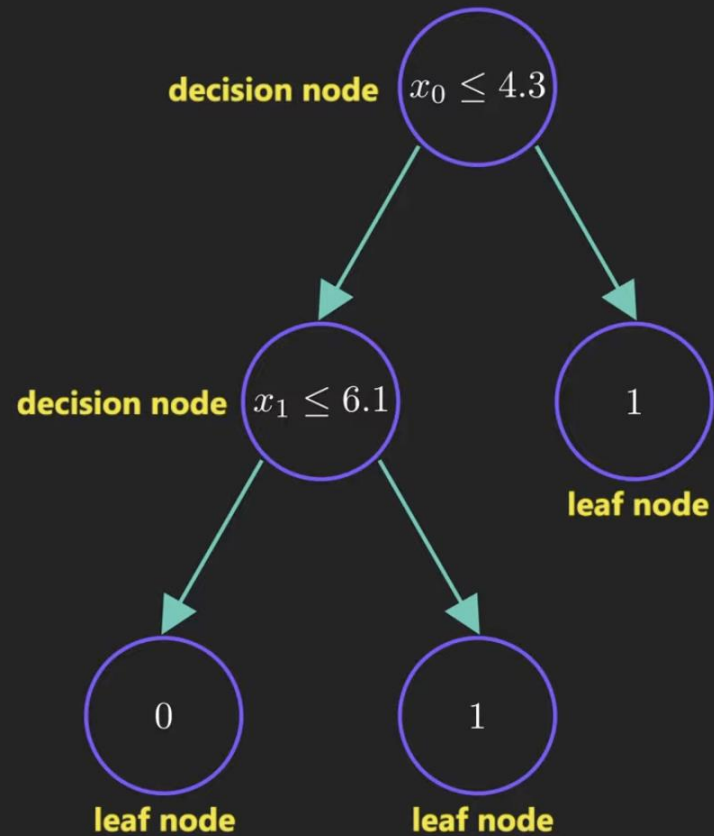
In random forests



Random Forest Algorithm

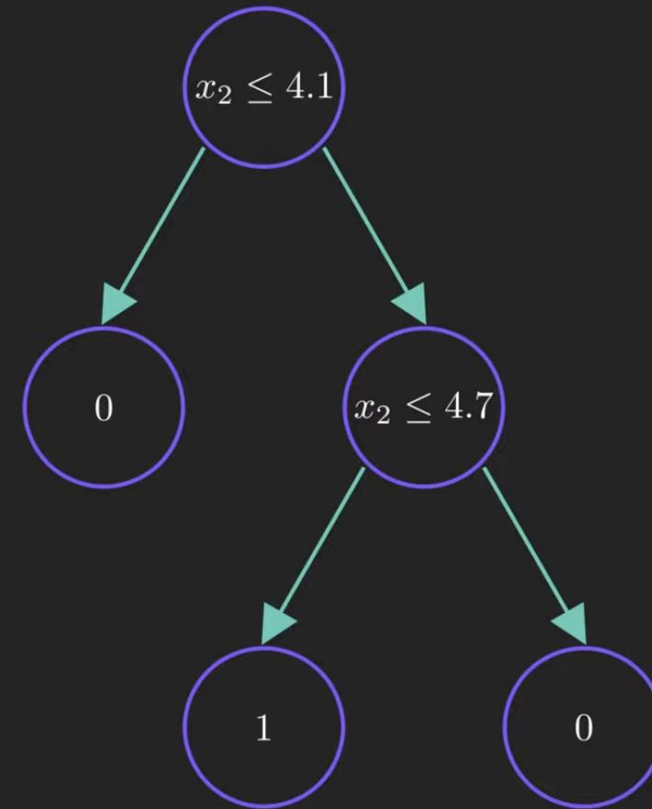
Why do we need RF while we have DT?

id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1



Lets change the data and see how the DT looks like

id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	6.5	4.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1



Random Forest

- We will build 4 datasets using bootstrap resampling with replacement. In practice, 100s of new bootstrapped datasets are created.
- Every dataset will contain the same number of rows as the original one.

<i>id</i>	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

<i>id</i>	<i>id</i>	<i>id</i>	<i>id</i>
2	2	4	3
0	1	1	3
2	3	3	2
4	1	0	5
5	4	0	1
5	4	2	2

Bootstrapped Datasets

Lets create decision trees for all bootstrapped datasets

- We will use a randomly selected number of feature subsets for each datasets.

<i>id</i>	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

<i>id</i>
2
0
2
4
5
5

x_0, x_1

<i>id</i>
2
1
3
1
4
4

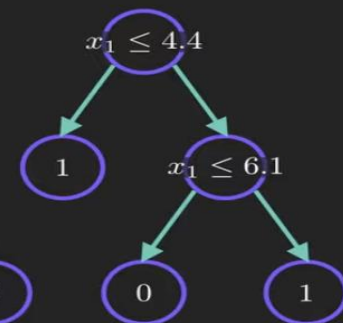
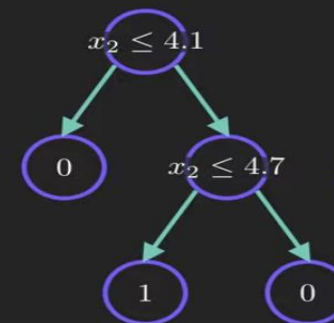
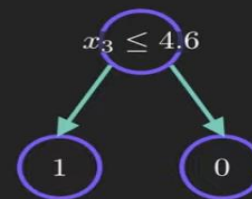
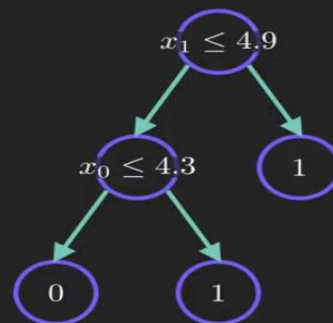
x_2, x_3

<i>id</i>
4
1
3
0
0
2

x_2, x_4

<i>id</i>
3
3
2
5
1
2

x_1, x_3



Lets see what happens with a new sample prediction

<i>id</i>	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

<i>id</i>
2
0
2
4
5
5

x_0, x_1

<i>id</i>
2
1
3
1
4
4

x_2, x_3

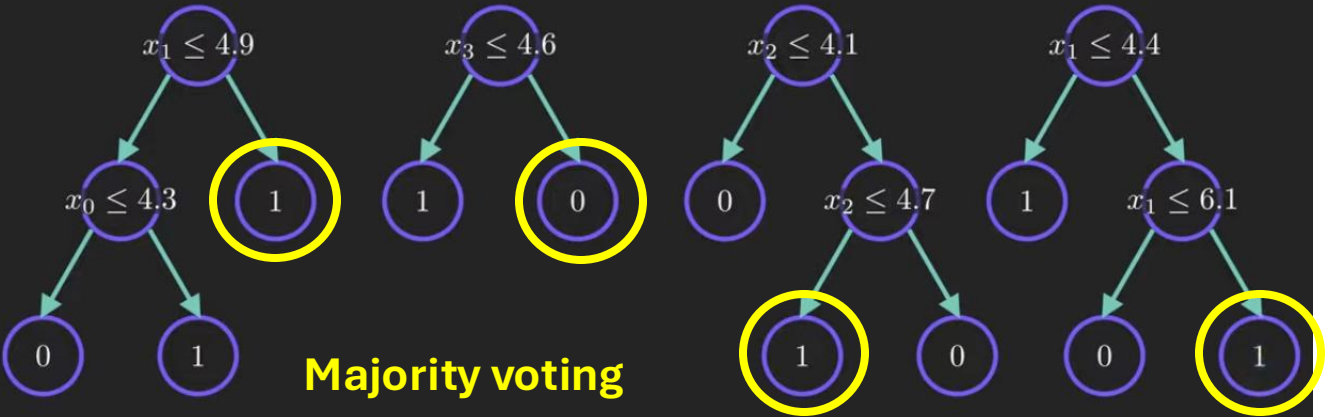
<i>id</i>
4
1
3
0
0
2

x_2, x_4

<i>id</i>
3
3
2
5
1
2

x_1, x_3

2.8	6.2	4.3	5.3	5.5
-----	-----	-----	-----	-----



**Bootstrap + Aggregation
(Bagging)**

Why it's called random forest?

- Because we have used two random processes, bootstrapping and random feature selection.

But what is the reason behind bootstrapping and feature selection?

- Well, bootstrapping ensures that we are not using the same data for every tree. So, in a way it helps our model to be less sensitive to the original training data.
- The random feature selection helps to reduce the correlation between the trees.
- If you use every feature, then most of your trees will have the same decision nodes and will act very similarly. That'll increase the variance.
- There's another benefit of random feature selection.
 - Some of the trees will be trained on less important features and they will give bad predictions. But there will also be some trees that give bad predictions in the opposite direction, so they will balance out.

Next point, what is the ideal size of the feature subset?

- Well, in our case we took 2 features which is close to the square root of the total number of features i.e. 5.
- Researchers found that values close to the log and sqrt of the total number of features work well.

How to use this for regression?

- While combining the predictions just take the average and you are all set to use it for a regression problem.



<https://www.youtube.com/watch?v=v6VJ2RO66Ag>